BIMM143 Bfactor Analysis

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```r
# Read PDB files
s1 <- read.pdb("4AKE")  # kinase with drug
```

```
##   Note: Accessing on-line PDB file
```

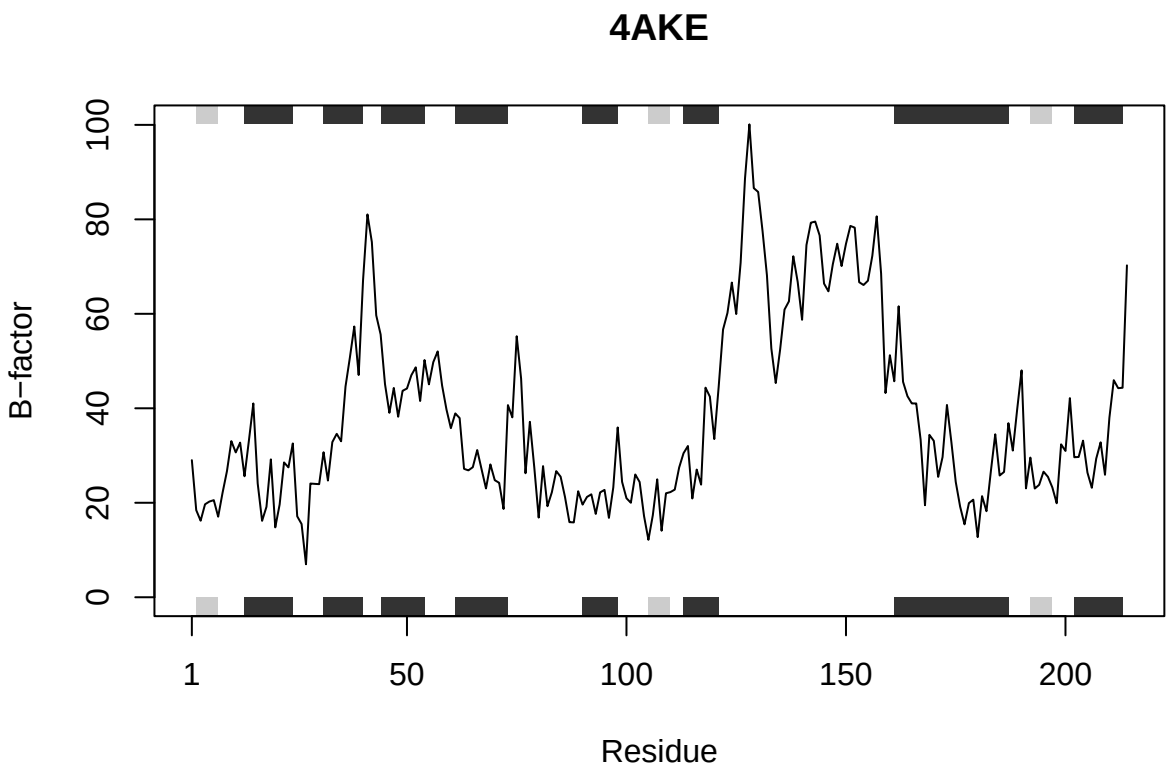
```r
s2 <- read.pdb("1AKE")  # kinase no drug
```

```
##   Note: Accessing on-line PDB file
##    PDB has ALT records, taking A only, rm.alt=TRUE
```

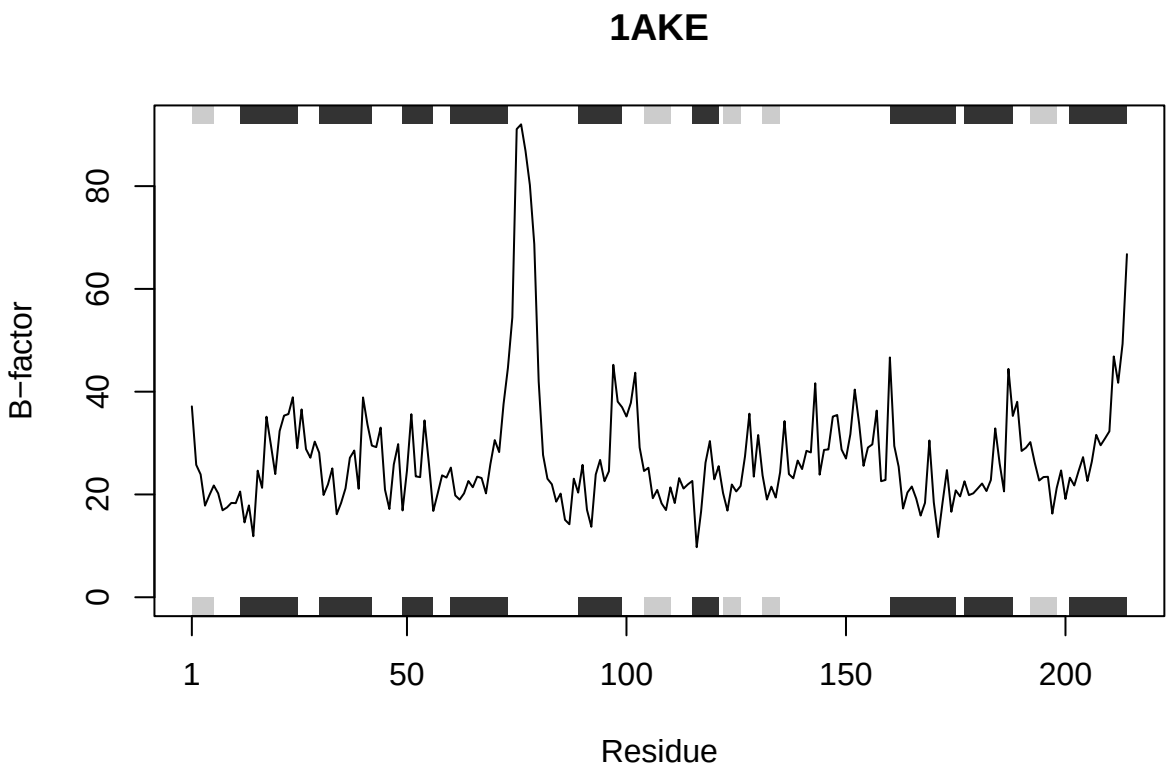
```r
s3 <- read.pdb("1E4Y")  # kinase with drug
```

```
##   Note: Accessing on-line PDB file
```

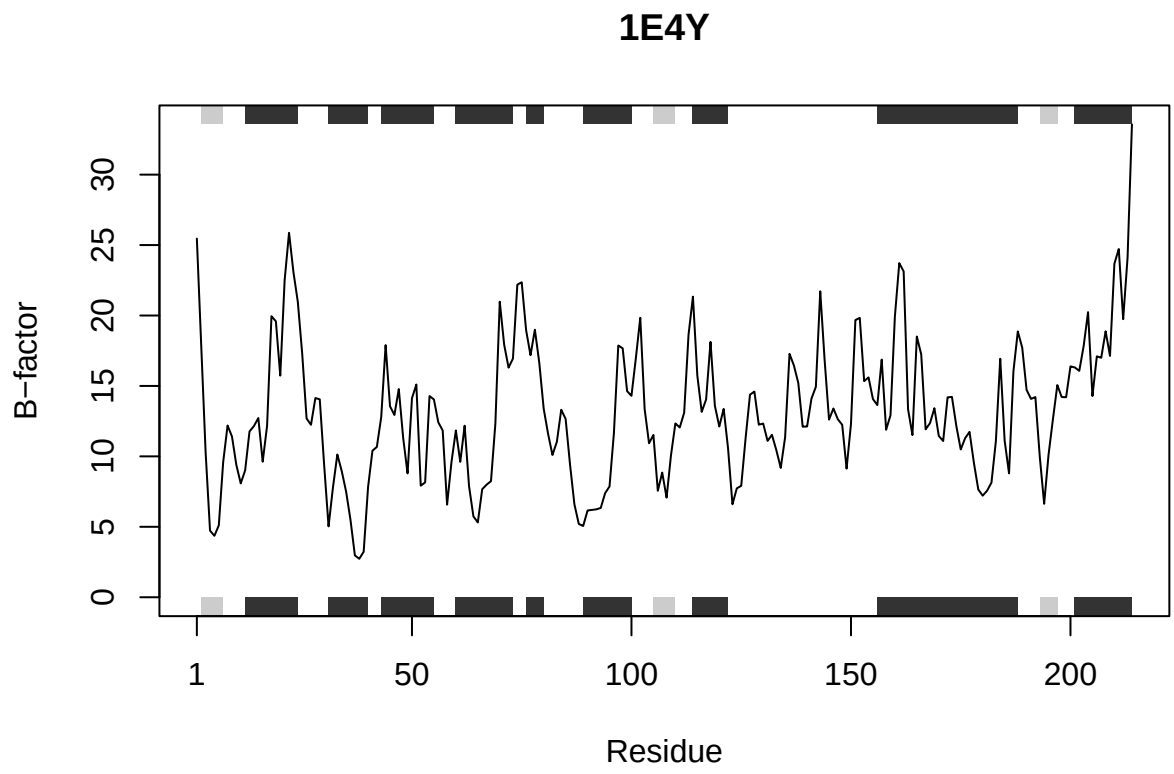
```r
# Trim to chain A and CA atoms
s1.chainA <- trim.pdb(s1, chain = "A", elety = "CA")
s2.chainA <- trim.pdb(s2, chain = "A", elety = "CA")
s3.chainA <- trim.pdb(s3, chain = "A", elety = "CA")

# Extract B-factors
s1.b <- s1.chainA$atom$b
s2.b <- s2.chainA$atom$b
s3.b <- s3.chainA$atom$b

# Plot B-factors
plotb3(s1.b, sse = s1.chainA, typ = "l", ylab = "B-factor", main = "4AKE")
```



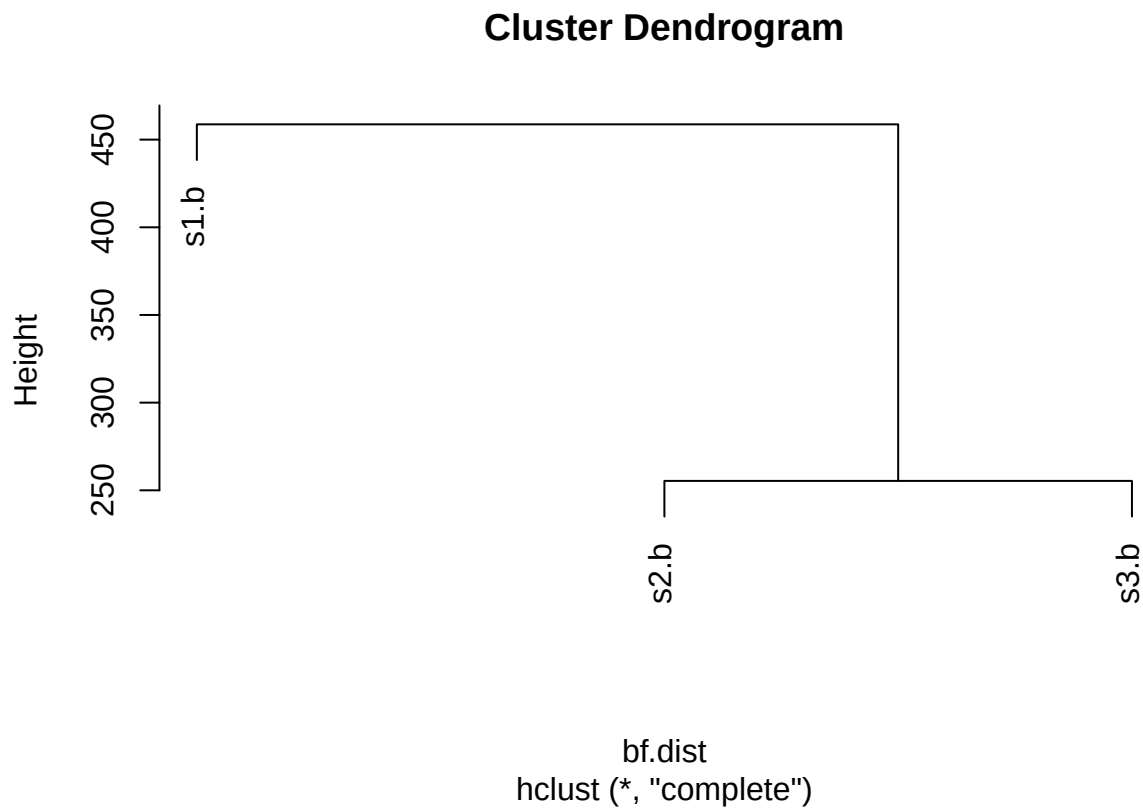

```
plotb3(s2.b, sse = s2.chainA, typ = "l", ylab = "B-factor", main = "1AKE")
```




```
plotb3(s3.b, sse = s3.chainA, typ = "l", ylab = "B-factor", main = "1E4Y")
```




```r
# Cluster analysis
bf.mat <- rbind(s1.b, s2.b, s3.b)
bf.dist <- dist(bf.mat)
bf.hc <- hclust(bf.dist)
plot(bf.hc)
```



Q1. What type of object is returned from the read.pdb() function?

A PDB object (a list-like R object) that stores the protein structure data (atoms, coordinates, etc.).

Q2. What does the trim.pdb() function do?

It cuts down the PDB to only the parts we want (ex: only chain A and only CA atoms).

Q3. What input parameter would turn off the marginal black and grey rectangles in the plots and what do they represent in this case?

They are secondary structure (helices/sheets) shown along the protein.To turn them off, we need to turn off the SSE display in plotb3()

Q4. What would be a better plot to compare across the different proteins?

Put all proteins on one plot (overlay the B-factor lines) so we can compare directly.

Q5. Which proteins are more similar to each other in their B-factor trends. How could you quantify this?

s1 and s2 are most similar (they cluster together first on dendrogram). we Quantify it by using dist() on the B-factor vectors and hclust() to cluster them. rbind(): Stacks vectors or rows together into a matrix or in our case puts all B-factor vectors into one object. dist(): Measures how different things are or in our case measures how different B-factor trends are between proteins. hclust(): Groups similar things together using distances or in our case groups proteins with similar B-factor patterns.

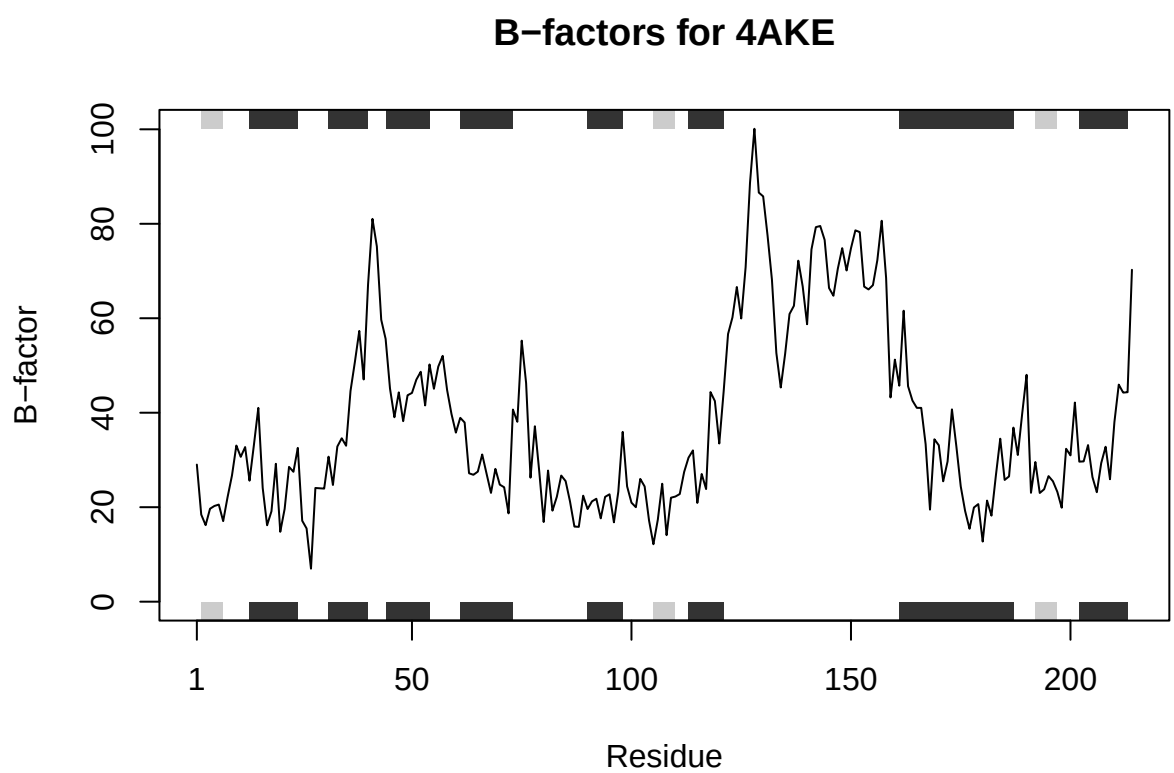
Q6. How would you generalize the original code above to work with any set of input protein structures?

The code can be generalized by writing a function that takes a PDB ID as input, trims the structure, extracts B-factors, and plots them automatically.

```
analyze_pdb <- function(pdb_id, chain = "A", elety = "CA") {
  # Inputs:
  # pdb_id: PDB ID string (e.g., "4AKE")
  # chain: which chain to keep (default "A")
  # elety: which atom type to keep (default "CA")

  pdb <- read.pdb(pdb_id)
  pdb_trim <- trim.pdb(pdb, chain = chain, elety = elety)
  b_factors <- pdb_trim$atom$b

  plotb3(b_factors,
         sse = pdb_trim,
         typ = "l",
         ylab = "B-factor",
         main = paste("B-factors for", pdb_id))

  return(b_factors)
}
```

```
bf_example <- analyze_pdb("4AKE")
```

```
##   Note: Accessing on-line PDB file
```



```r
head(bf_example)
```

```
## [1] 29.02 18.44 16.20 19.67 20.26 20.55
```